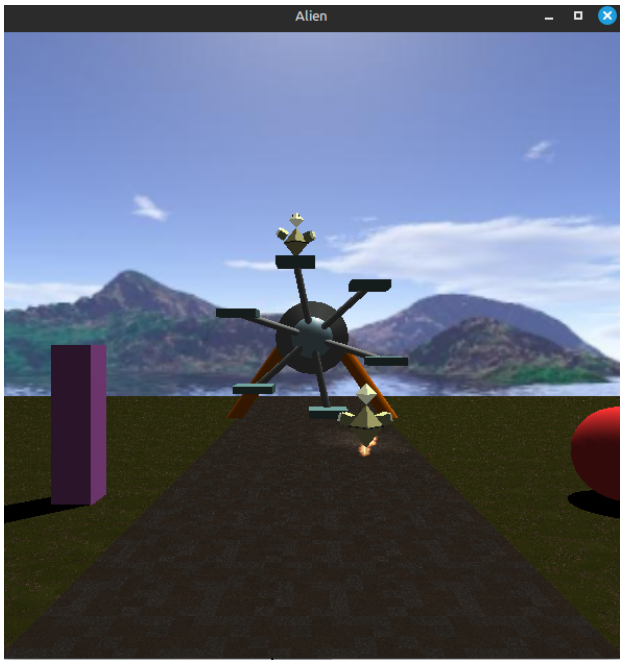
Jamie Boyd
SN: 67354831

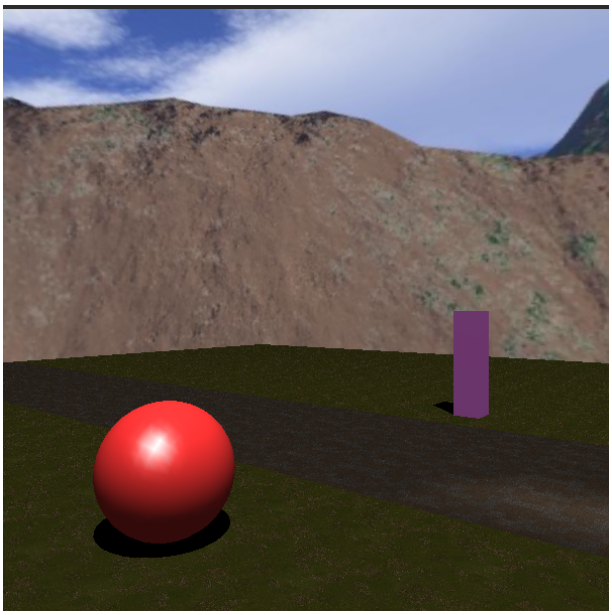
My scene contains a ferris wheel, two aliens, a floor, a skybox, a pole and a sphere. The ferris wheel is spinning with one alien on it that is constantly waving its arms, and also stops to let the alien get on and off the ferris wheel. The other alien is busy pacing back and forth along the ground, with a spotlight coming from his eyes to help him see. Both aliens have fire shooting towards the ground which allows them to navigate, as opposed to having legs. The pole and sphere both have a shadow.



Here you can see the alien waiting on the ground for the ferris wheel, using his fire for propulsion, along with the other alien pacing towards the camera with its spotlight in front of it.

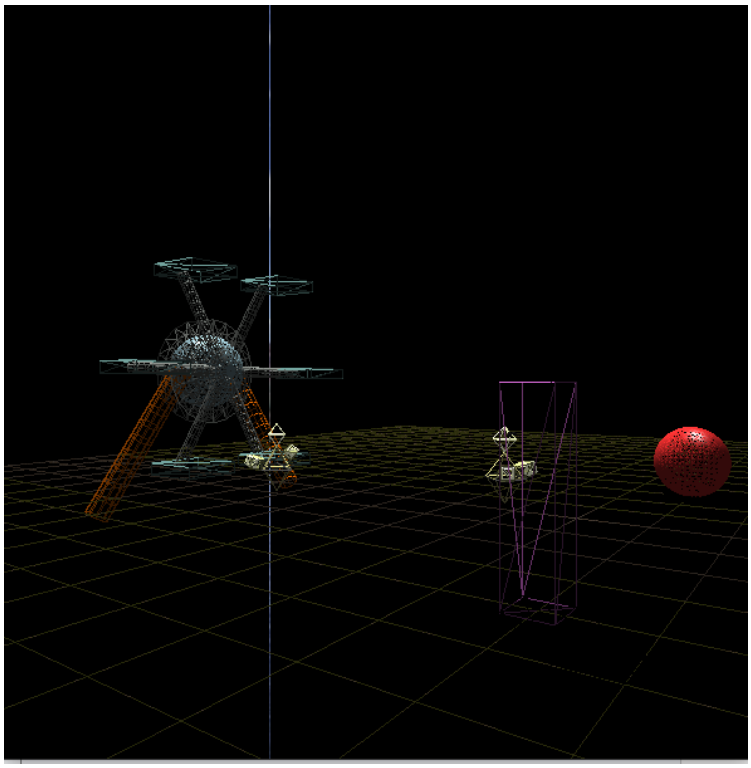


Here is the alien on the ferris wheel, where his fire is turned off because he is sitting on the seat, along with the other alien pacing away from the camera, where its spotlight is still facing the same direction as its eyes.



Here you can see another angle of the shadows that are cast by the pillar and sphere.

Here is a screenshot with wireframe mode on.



Extra Features

- Planar shadows from the pillar and sphere
- Aliens get on and off the ferris wheel
- Sky box
- Dynamic spotlight that moves with the alien pacing on the ground
- Smooth user movement
- Particle system which is the fire shooting beneath the aliens, that also turns off when they aren't moving (e.g On the ferris wheel).

Sketches

No diagrams or sketches were used for the design of models.

Equations

No equations were used to design surfaces or animations

Controls

- Up: Camera forwards
- Down: Camera backwards
- Left: Turn camera left
- Right: Turn camera right
- q: Toggle wireframe mode

Build Instructions

The compiling process is the same as that of the labs. Copy the files into vscodium, transfer the .tga files into the build folder that is created by cmake, then set the .cpp file as the build/launch target to run it using cmake.

References

Fire particle texture, created by user bart under the CC0 public domain license.
https://opengameart.org/node/7704

Grass texture, author Amal Kumar. Licensed CC0 public domain.
https://polyhaven.com/a/rocky_terrain_02

Concrete texture, author Charlotte Baglioni. Licensed CC0 public domain.
https://polyhaven.com/a/concrete_pavement_03

Skybox texture, skybox5, given under section 4 of lecture material on learn.

GenAI Statement

Gen ai was used in this assignment to explain some computer graphics functions and answer questions about debugging.

Declaration

I declare that this assignment submission represents my own work (except for allowed material provided in the course), and that ideas or extracts from other sources are properly acknowledged in the report. I have not allowed anyone to copy my work with the intention of passing it off as their own work.

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Date: 30/03/2026